

Activity 7

Aim: To determine pH of NaOH solution of various concentrations by using pH paper or universal indicator.

Apparatus: Test tubes, test tube stand.

Chemicals: 0.1 M NaOH, 0.01 M NaOH, 0.001 M NaOH solutions, pH paper or universal indicator.

Theory: $\text{pOH} = -\log_{10}[\text{OH}^-]$; $\text{pH} = 14 - \text{pOH}$

For Acidic solution $\text{pH} < 7$, Basic solution $\text{pH} > 7$, Neutral solution $\text{pH} = 7$.

The value of pH changes with change in concentration of NaOH.

Procedure:

1. Take 5 ml each of 0.1M, 0.01M, 0.001M NaOH solutions in separate test tubes marked A, B, C.
2. Add 2 drops of universal indicator in each test tube or put piece of pH paper in each test tube.
3. Observe the Colors of solution or pH paper and record pH of each solution.

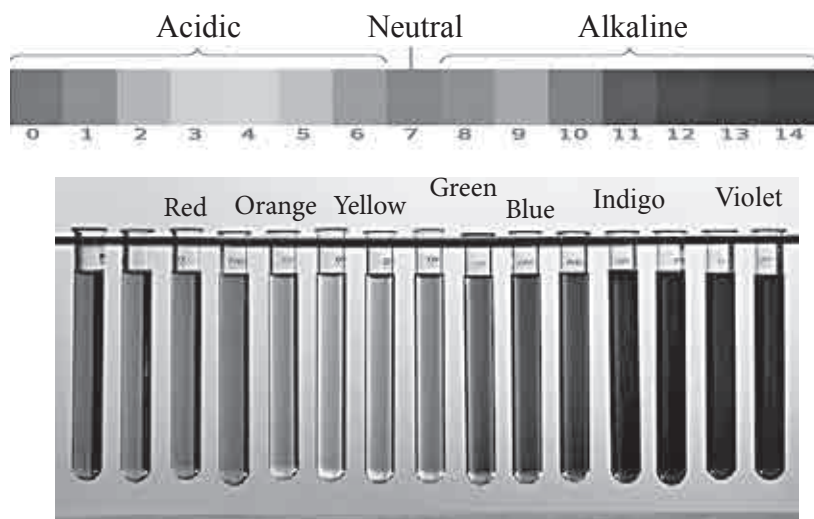


Fig. Activity 7.1 pH range

Note: Student has to prepare 0.01M and 0.001M NaOH from the given 0.1M NaOH solution.

Observation Table/ Conclusion:

Sr. No.	Solutions	Concentration of solutions	Color of pH paper or solution	pH
1	A	0.1 M NaOH	Violet	13-14
2	B	0.01 M NaOH	Indigo	12
3	C	0.001 M NaOH	Blue	10-11

Remark and sign of teacher:

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

- If nature of the solution is alkaline is then its pH values is
 - ☒ a. > 7
 - b. < 7
 - c. Equal to 7
 - d. In between 7 to 14
- Which of the following indicator have different pH range is
 - a. Phenolphthalein indicator
 - b. Acid-base indicator
 - c. Methyl orange indicator
 - ☒ d. Universal indicator
- The negative logarithm to the base 10 of OH^- ions concentration is known as
 - ☒ a. pH
 - b. pOH
 - c. pKa
 - d. pKb
- The color of universal indicator in strongly basic solution is
 - a. red
 - ☒ b. violet
 - c. orange
 - d. blue
- If concentration of OH^- ions increase, the values of the pH of the solution will be
 - a. decrease
 - b. remain constant
 - ☒ c. increase
 - d. first increases then decreases

Short answer questions

1. What is pOH of the solution?

Ans. pOH of a solution is defined as the negative logarithm (base 10) of the concentration of OH ions in the solution.

2. If the concentration of given solution of KOH is 0.01M. Calculate the pOH and pH of the solution. $pOH = -\log(0.01)$

Ans. $\text{pOH} = -\log(0.01)$
 $\text{pOH} = 2$
 $\text{pH} + \text{pOH} = 14$
 $\text{pH} = 14 - 2 = 12$
 Thus, $\text{pH} = 12$
 and $\text{pOH} = 2$.

Remark and sign of teacher: